

Syllabus

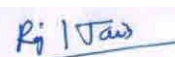
UG0802 -ZOO-76T-303- Microbiology, Immunology & Biostatistics

UG0802 -ZOO-76P-304- Practicals based on Microbiology, Immunology & Biostatistics

VI-Semester-Zoology

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
VI	ZOO-76T-303 ZOO-76P-304	Microbiology, Immunology & Biostatistics Practicals based on Microbiology, Immunology & Biostatistics			7	6
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
7	MJR	4	2	6	Yes	Lectures: 60 lectures including diagnostic and informative assessments during lecture hours and 30 Hours of Practical training/demonstration.
List of Programme Codes in which Offered as Minor Discipline		B.Sc. Chemistry: UG0804 B.Sc. Botany: UG0805				
Prerequisites		V Semester (B.Sc. Bio group)				
Objectives of the Course		- The objectives of studying a combined course in Microbiology, Immunology, and Biostatistics are to develop a foundational understanding of microorganisms, the immune system, and statistical methods. - Students will acquire practical skills in microbiological and immunological techniques, including culturing, staining, and identifying microbes, as well as conducting immunoassays. - The course aims to help students understand the interplay between microbiology, immunology, and biostatistics, applying statistical analysis to interpret experimental data in these fields. - It enhances abilities in designing experiments, collecting and analyzing data, and drawing valid conclusions, thereby fostering research capabilities. - The course prepares students for careers in biomedical research, clinical diagnostics, public health, and related fields by understanding the applications of these disciplines in health and disease.				

Detailed Syllabus


 Dy. Registrar
 (Academic)
 University of Rajasthan
 JAIPUR

ZOO- 76T-303: Microbiology, Immunology & Biostatistics

Unit-I

History of Microbiology: Contributions of Anton van Leeuwenhoek, Louis Pasteur, Edward Jenner and Robert Koch. **2 Hrs**

Basic structural and functional characteristics of bacteria, fungus, cyanobacteria, rickettsiae, chlamydiae, mycoplasmas and virus. Structural organization of bacteria- flagella, capsule, cell wall, cell membrane, cytoplasm, chromosome, growth phases **4Hrs**

Techniques in Microbiology: pure culture technique, growth conditions (temperature, pH, nutrients), Gram's staining, fungal staining **3 Hrs**

Bacterial diseases: causative agent, brief pathology, treatment, prevention of Diphtheria, meningitis, tuberculosis, botulism, tetanus, leprosy **3 Hrs**

Viral diseases: causative agent, brief pathology, treatment, prevention of influenza, herpes, measles, chicken pox, small pox, rabies, polio, AIDS, covid-19. **3 Hrs**

Unit-II

Components of immune system: different types of cells and organs involved. **3 Hrs**

Antigens: Essential features of antigen, antigenic determinants, Immunological valence, adjuvants, haptens **2 Hrs**

Antibodies: Nature and Primary structure of immunoglobulins; Types of immunoglobulins: IgG, IgM, IgA, IgD and IgE and their structure and functions **3Hrs**

Antigen Antibody reactions: Precipitation and agglutination **1Hrs**

Types of immunity: Active and passive immunity, Cell mediated immunity, humoral immunity, Auto immunity **3Hrs**

Basic idea of interferon, Complement system and MHC **3Hrs**

Unit III

Application of microbes: Use of microbes in fermented foods, dairy and beverages: wine, beer, cheese, curd and vinegar **5Hrs**

Use of microbes in waste water treatment, bioremediation (brief idea) **4 Hrs**

Types of vaccines: Live attenuated, inactivated vaccine, DNA and recombinant vaccine **3 Hrs**

Monoclonal antibodies production and application, Abzymes **3 Hrs**

Unit-IV

Biostatistics and its applications, variable vs attributes, population vs sample, arrangement of data, frequency distribution **3 Hrs**

Graphical representation of data: line diagram, bar diagram, pie chart, histogram and polygon **4 Hrs**

Measures of central tendency: arithmetic mean, median, mode **4Hrs**

Standard deviation, standard error of mean **4Hrs**

Suggested Books and References:

1. Fundamentals of microbiology by Edward Alcamo.Benjamin/Cummings publishing company,Inc.
2. Prescott's Microbiology by Joanne.M.Willey,LindaSherwood,Christoppher J. Woolverton,2017. McGraw-Hill Education
3. Kindt, T. J., Goldsby, R. A., Osborne, B. A., & Kuby, J. (2007). *Kuby Immunology*(6th ed.). New York: W.H. Freeman and Company.
4. Owen, J. A., Punt, J., Stranford, S. A., & Jones, P. P. (2018). *Kuby Immunology* (8th ed.). New York: W.H. Freeman and Company.
5. Abbas, A.K. & Lichtman, A.H. (2001). Basic Immunology: Functions and Disorders of Immune System. US: W.B. Saunders.
6. Pagano, M., & Gauvreau, K. (2018). Principles of Biostatistics (2nd ed.). CRC Press.
7. Rosner, B. (2015). Fundamentals of Biostatistics (8th ed.). Cengage Learning.
8. Microbiology -concepts and applications (2023) Micheal Joseph Pelczar,Eddie Chin Sun Chan,Noel R. Krieg. McGraw-Hill Education
9. Delves, P.J., Martin, S.J., Burton, D.R., &Roitt, I.M (2011). Roitt's Essential Immunology (12thed.). New Jersey, USA: John Wiley & Sons Ltd.

Suggested E-resources:

1. Microbiology Society: Offers educational resources, including articles and practical resources for learning microbiology.
2. OpenStax - Microbiology Textbook: A free, peer-reviewed textbook that covers a wide range of topics in microbiology.
3. MicrobeWiki: A student-edited resource that provides detailed information on various microorganisms.
4. Immunology Online Textbook: A comprehensive online textbook from NCBI that covers various aspects of immunology.
5. Janeway's Immunobiology: A highly regarded textbook available online that provides a deep dive into immunology.
6. Statistical Methods for the Analysis of Biomedical Data: An open-access book that focuses on statistical methods used in biostatistics.
7. OpenIntro Statistics: A free textbook that provides an introduction to statistics, including

applications in biostatistics.

Course Learning Outcomes: After successful completion of the course, students will be able to:

- Understand the basic concepts and applications of Microbiology, and Immunology and Biostatistics.
- Explain the structure and function of microorganisms.
- Describe the role of microbes in the environment, industry, and human health.
- Explain the principles and applications of immunological techniques.
- Apply appropriate statistical methods to analyze biological data, interpret results accurately, and make informed decisions based on statistical evidence.
- Gain proficiency in using statistical software for data management, analysis, and visualization, enabling them to handle complex datasets effectively in biomedical research contexts.

Practical Syllabus

ZOO-76P-304- Practicals based on Microbiology, Immunology & Biostatistics

Microbiology

1. Preparation and use of culture media for microbes (NA and PDA).
2. Culture of microbes (bacteria and fungi) from food (curd/ spoiled food).
3. Gram's staining for bacteria.
4. Simple staining of bacteria.
5. Slide preparation and identification of any two fungi.

Immunology

6. Tests of blood grouping.
7. Widal test to identify any pathogen in blood.
8. Differential leucocyte count (DLC) in the blood sample.

Biostatistics

9. Representation of data by using bar diagram and pie charts.
10. Numericals based on biostatistical measurement: Mean, Mode and Median.

11. Numericals based on standard deviation.
12. Educational trip to any Microbiology Lab/ Dairy/industry.

Scheme of Practical Examination and Distribution of Marks

S.No.	Practical Exercises	Regular Students	Ex. /N.C. Students
1.	Major exercise (Microbiology)	8	12
2.	Minor exercise (Immunology)	7	6
3.	Minor Exercise (Biostatistics)	7	6
3.	Identification and comments on Spots (1 to 8)	16	16
4.	Viva Voce	7	10
5.	Class Record	5	
		50	50

Note:

*Internal marks for regular students only.

1. In case of non-availability, the exercise should be substituted with diagrams/photographs.
2. Candidates must keep a record of all work done in the practical class and submit the same for inspection at the time of the practical examination.
3. Mounting materials for permanent preparation will be made available according to the course or through collection and preservation methods.
4. Computer-aided techniques should be adopted as per UGC guidelines.